Grade 11 · Algebra II · Topic 6: Exponential and Logarithmic Functions · Lesson 6-7 — Geometric Sequences and Series

Choose Your Route

Three routes, same destination. Start anywhere — you may switch routes at any time, and you may try more than one.

Student name :	Date : .....

PRACTICE — Secure the method

Find it, write it, add it

Items 1–3: answer, and write which operation repeats. Items 4–6 are below.

1.	2.	3.
$2,\ 10,\ 50,\ \ldots:$ find r	$a_1 = 6,\ r = 2:$ find a_5	$1 + 3 + 9 + 27$

4. Write the recursive and explicit formulas for $100,\ 50,\ 25,\ \cdots$, and find the sixth term.

5. Evaluate $\sum_{k=1}^{6} 3 \cdot 2^{k-1}$. Name the first term, ratio and number of terms first.

6. Find the sum of the first 5 terms of $4 - 8 + 16 - \cdots$.

Done when: the ratio is found by dividing consecutive terms, and each formula reproduces the first terms when substituted.

APPLY — Use it in context

Work back, model it, sum it

1. A geometric sequence has $a_3 = 45$ and $a_5 = 405$. Find both possible sequences, and write their first five terms.

2. A model Jeddah electric-bus programme (model values only) adds 32 buses in year 1, and each year 1.5 times as many as the year before. Find the number added in year 6 and the total after 6 years.

3. Evaluate $\sum_{k=1}^{7} 6 \cdot (-2)^{k-1}$. Check by writing out the seven terms.

Done when: the number of steps between terms is shown, and each sum names its first term, ratio and number of terms.

INVESTIGATE — Find out why

Arguments, not answers

1. A student says the sum of the first 10 terms of $1,\ 2,\ 4,\ \cdots$ is $2^{10} = 1024$. Find the error and the correct sum.

2. Find $S_1,\ S_2,\ S_3,\ S_4$ for $1 + \frac{1}{3} + \frac{1}{9} + \cdots$. What value do they seem to approach? Use the formula to support your answer.

3. Let $S_4 = 3 + 3r + 3r^2 + 3r^3$. Multiply by r, subtract, and show that $S_4(1 - r) = 3\,(1 - r^4)$.

Done when: each explanation uses the constant ratio or the cancelling terms, and the exact value is given before any decimal.

Remember

These are names for the tasks, not names for you. Move between them whenever you want to. Everyone answers the same “Time to Check” at the end of the lesson.